



Effect of microalloying of Nb on corrosion resistance and thermal stability of ZrCu-based bulk metallic glasses

X.P. Nie^a, X.M. Xu^b, Q.K. Jiang^a, L.Y. Chen^a, Y. Xu^a, Y.Z. Fang^c, G.Q. Xie^d,
M.F. Luo^d, F.M. Wu^c, X.D. Wang^a, Q.P. Cao^a, J.Z. Jiang^{a,*}

^aInternational Center for New-Structured Materials (ICNSM), Zhejiang University and Laboratory of New-Structured Materials, Department of Materials Science and Engineering, Zhejiang University, Hangzhou 310027, PR China

^bHospital of Zhejiang University, Zhejiang University, Hangzhou 310027, PR China

^cDepartment of Physics, College of Mathematics and Physics, Zhejiang Normal University, Jinhua, Zhejiang 321004, PR China

^dZhejiang Key Laboratory for Reactive Chemistry on Solid Surfaces, Institute of Physical Chemistry, Zhejiang Normal University, Jinhua, Zhejiang 321004, PR China

ARTICLE INFO

Article history:

Received 27 April 2008

Received in revised form 9 October 2008

Available online 4 December 2008

PACS:

71.23Cq

81.05.Kf

Keywords:

Metallic glass

Corrosion resistance

Thermal stability

Glass forming ability

ABSTRACT

Bulk metallic glasses (BMGs) of $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ with $x = 0, 0.5, 1, 2$ and 4 at.% were prepared by copper mould casting. The corrosion resistance of the ZrCu-based BMGs with different Nb contents was carefully examined by weight loss measurements and potentiodynamic polarization tests in 3 mass% NaCl, 1 N HCl and 1 N H_2SO_4 solutions, respectively. Nb addition improves the newly developed BMGs' corrosion resistance in chloride-containing solutions and the alloys all exhibit excellent corrosion resistance in 1 N H_2SO_4 . The corrosion behavior of the alloy containing 0 and 4 at.% Nb in phosphate-buffered solution was examined by electrochemical polarization tests. The influence of Nb addition on glass forming ability (GFA), thermal stability and mechanical property was investigated by X-ray diffraction, differential scanning calorimetry and compression tests, respectively. It is found that the addition of Nb can deteriorate the GFA and thermal stability of the base system, but little effect is observed on the mechanical properties, e.g., yielding strength and plasticity, of the ZrCu-based BMG alloys.

© 2008 Elsevier B.V. All rights reserved.

1. Introduction

Since last decade, considerable attention has been focused on the bulk metallic glasses (BMGs) due to their unique properties [1–3]. A series of BMG-forming systems with critical diameters more than 20 mm have been developed. They are Zr- [4], Pd- [5], Y- [6], Pt- [7], Mg- [8], La- [9] and ZrCu-based [10] alloys. Among them, ZrCu-based alloys are considered to be a promising system because it exhibits (1) excellent glass forming ability (GFA) up to 25 mm BMG prepared by copper mould injection; (2) excellent mechanical properties: yield strength 1822 MPa, fracture strength 2163 MPa, Vicker's hardness 533 ± 3 ; (3) large compressive plastic deformation up to 15%; (4) relative cheap raw materials (no expensive Pd, Pt and Sc elements); (5) non-toxic elements (e.g., Be); (6) good potential biocompatible property (no Ni element) [11]. To have industrial applications of the novel ZrCu-base BMG, one has to address the life-time (corrosion behavior) of the BMG under various environmental conditions. In our preliminary corrosion study [11] of the ZrCu-based BMGs, it was found that the BMG alloy has

high corrosion resistance in H_2SO_4 solution, but poor corrosion resistance in 3% NaCl and 1 N HCl. Thus, it is of interest to improve corrosion resistance of the BMG alloy in chloride-containing solutions and understanding the pitting mechanism.

Several methods have been reported to improve BMGs' corrosion resistance: (1) changing the surface composition or structure through micro-arc oxidation, ion-implantation or any other surface treatment methods [12,13]; (2) decreasing the free volume, by which the corrosion resistance of Zr-based alloys can be improved [14]; (3) forming glass-matrix composite [15], i.e. introducing second phase into the amorphous matrix either by annealing or in situ synthesis; (4) adding elements having strong passivation ability into the base alloy system [16,17]. In this work, the influence of Nb addition on corrosion resistance in 3% NaCl, 1 N HCl, 1 N H_2SO_4 and phosphate-buffered solution, glass-forming ability, thermal stability and mechanical property of $\text{Zr}_{46}\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ BMG has been investigated. The pitting mechanism of Nb-bearing ZrCu-based BMG alloy is tentatively proposed. The reasons for choosing Nb as the additional element are as follows: (1) as a valve-metal Nb has strong passivation ability, and (2) it is reported that refractory metals such as Nb and Ta can effectively enhance the plasticity of Zr-based BMGs by microalloying [18,19].

* Corresponding author.

E-mail address: jiangjz@zju.edu.cn (J.Z. Jiang).

2. Experimental

Master alloy ingots with nominal compositions of $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ with $x = 0, 0.5, 1, 2$ and 4 at.% were prepared by arc-melting the mixture of pure Zr (99.8 wt.%), Cu (99.9 wt.%), Ag (99.9 wt.%), Al (99.99 wt.%) and Nb (99.5 wt.%) metals under a Ti-gettered high purity argon atmosphere. Each ingot was remelted at least four times to make sure its chemical homogeneity. Cylindrical glass rods with a diameter of 2, 3, 8, 10, 12 and 20 mm were prepared by injection casting in a high-vacuum (2×10^{-5} Torr) chamber under a purified argon atmosphere, in which 20 mm is the maximum size of copper mold for casting in our laboratory. The transverse cross-sections of the samples were examined by X-ray diffraction (XRD) on a Thermo ARL X'Tra diffractometer with $\text{Cu K}\alpha$ radiation at the operating voltage 45 kV so as to examine the structure of the samples prepared. The thermal response of each composition was analyzed on a NETZSCH DSC 404 C differential scanning calorimeter under a continuous argon flow at a heating rate of 0.33 K/s. Cylindrical sample rods (2 mm in diameter and 4 mm in height) were used for compressive tests, which were performed using a calibrated universal testing machine (CMT5205 SANS, China) at a strain rate of $4 \times 10^{-4} \text{ s}^{-1}$. Three samples were measured to ensure reproducible results.

Corrosion behaviors of the metallic glasses were evaluated by immersion tests and potentiodynamic polarization measurements. Electrolytes of 1 N HCl, 3 mass% NaCl and 1 N H_2SO_4 aqueous solutions prepared from reagent grade chemicals and deionized water were used at room temperature (about 300 K) open to air. Corrosion-penetration rate (CPR) was estimated from the weight loss after immersion in the undisturbed solutions for durations from 48 to 336 h. Electrochemical measurements were conducted in a three-electrode cell using a platinum counter electrode and a standard saturated calomel reference electrode (SCE). Potentiodynamic polarization curves were recorded at a potential sweep rate of 2 mV/s after immersing the specimens for about 20 min when the open circuit potentials became almost steady. The electrochemical tests in phosphate-buffered solution (PBS, pH 7.42) were performed at 37 °C. The electrolyte is composed of 8.01 g/L NaCl, 0.20 g/L KCl, 2.90 g/L $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$ and 0.20 g/L KH_2PO_4 . The dissolved-oxygen content of this electrode was reduced to a phys-

iologically relevant value of $3.3 \times 10^{-2} \text{ mol/m}^3$ by aerating the solution with a 4% O_2/N_2 gas mixture at a flow rate of 50 ml/min. One hour before each test, aeration was performed and continued at this rate throughout the tests. Prior to the tests, the specimens were all mechanically polished to mirror finish with continuous SiC paper up to 1200 grit and then degreased in acetone, washed in distilled water, dried in air and further exposed to air for about 24 h before the tests. In order to get some information on the chloride-induced localized corrosion mechanism, some selected specimens after polarization tests in 3% NaCl were immediately removed, cleaned in ethanol and examined by scanning electron microscopy (HITACHI S-4800) equipped with energy-disperse X-ray (EDX) facility.

3. Results

3.1. Corrosion measurements

The average corrosion rates of the as-cast $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ ($x = 0, 0.5, 1, 2$ and 4 at.%) alloys and AISI stainless steel 304L immersed in 3 mass% NaCl, 1 N HCl and 1 N H_2SO_4 solutions at 298 K open to air for 48 to 336 h are shown in Fig. 1(a)–(c). It is seen from Fig. 1(a) that the corrosion rates of the BMGs immersed in 3% NaCl solutions show a vivid step-like shape, i.e. the more Nb content, the lower average corrosion rate. The alloy with 4 at.% Nb shows the lowest corrosion rate of $0.012 \text{ mm year}^{-1}$, which is only about 10% of the Nb-free alloy and one third of steel. When immersed in 1 N HCl (see Fig. 1(b)), it also appears that the average corrosion rate decreases with Nb addition. The corrosion rate for the Nb-free alloy is approximately three times of the alloy containing 4 at.% Nb. In the H_2SO_4 solutions (Fig. 1(c)), the BMGs exhibit superior corrosion resistance and the average corrosion rates are all below $0.010 \text{ mm year}^{-1}$. In contrast much higher corrosion rate of $0.092 \text{ mm year}^{-1}$ is exhibited by steel, which is being about more than one order of magnitude higher than the BMGs in 1 N H_2SO_4 . Thus, it is clear that in all solutions examined, tiny Nb addition is effective to enhance corrosion resistance of the novel ZrCu-based BMGs while the BMG with 4 at.% of Nb exhibits the lowest corrosion rates among the other BMGs.

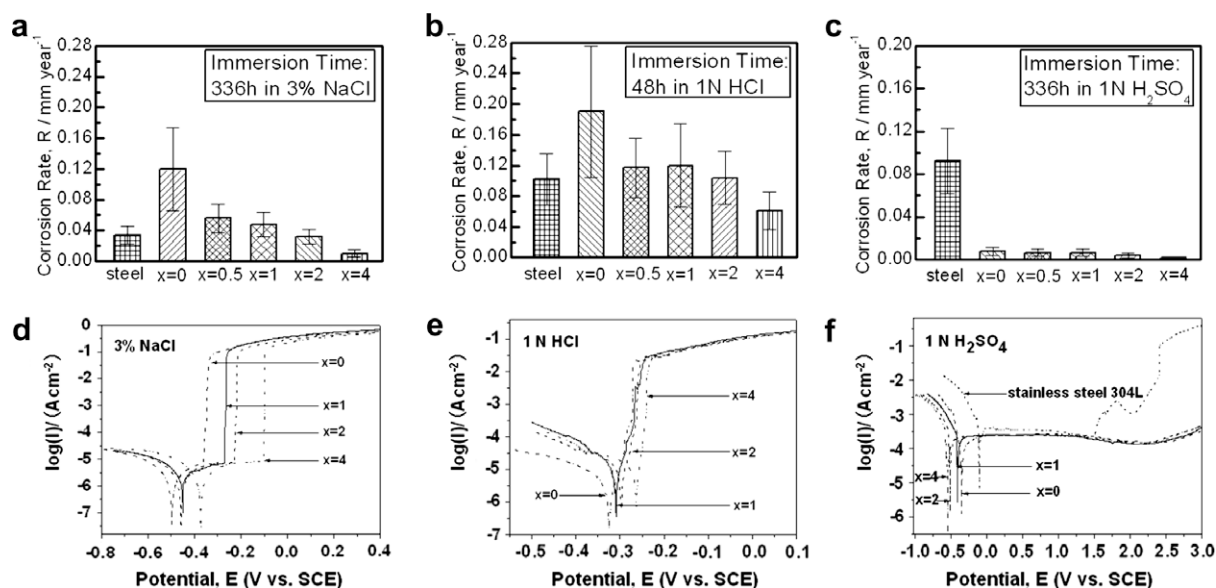


Fig. 1. Average corrosion rates (a–c) of $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ ($x = 0, 0.5, 1, 2$ and 4 at.%) alloys and AISI stainless steel 304L in 3% NaCl solution (a), 1 N HCl solution (b), and 1 N H_2SO_4 solution (c). Potentiodynamic curves (d–f) of $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ ($x = 0, 1, 2$ and 4 at.%) alloys in 3% NaCl solution (a), 1 N HCl solution (b), and 1 N H_2SO_4 solution (c) together with AISI stainless steel 304L for comparison.

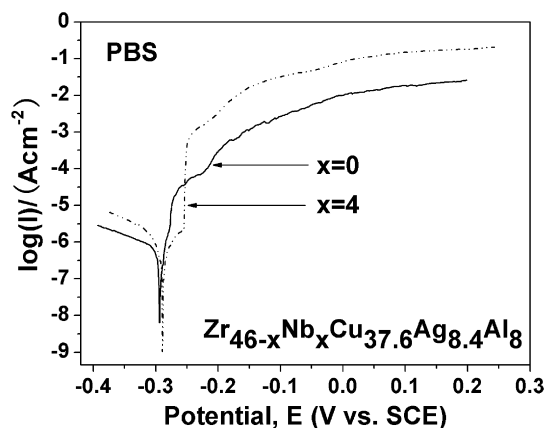


Fig. 2. Potentiodynamic polarization curves for the $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ ($x = 0$ and 4 at.%) alloys in phosphate-buffered solutions at 37 °C.

Fig. 1(d)–(f) show potentiodynamic polarization measurements for $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ ($x = 0, 1, 2$ and 4 at.%) BMG alloys in the above-mentioned three electrolytes open to air at room temperature. In 3% NaCl solution (Fig. 1(d)), all of the alloys were spontaneously passivated with low passive current densities of about $8 \mu\text{A cm}^{-2}$. The difference, $|E_{\text{pit}} - E_{\text{corr}}|$ also increases from 0.13 to 0.27 V with Nb content from 0 to 4 at.%. This demonstrates that highly protective surface films can be immediately formed in 3% NaCl solution by anodic passivation and the pitting susceptibility is largely enhanced by Nb addition. We also conducted polarization tests for the sample containing 8 at.% Nb. No obvious difference was detected as compared to the sample with 4 at.% Nb. In 1 N HCl solution (Fig. 1(e)) all the alloys dissolve quickly by slight anodic polarization and no self-passivation is revealed. This indicates that the unstable surface films of the alloys formed in solutions

have limit resistance to the joint attack of H^+ and Cl^- ions. Thus, the newly developed ZrCu-based alloys show relatively inferior corrosion resistance in 1 N HCl although Nb addition slightly shifts corrosion potential to positive direction. In contrast, the BMGs exhibit fully different corrosion behaviors in 1 N H_2SO_4 solutions (see Fig. 1(f)), which reveal excellent passivation behavior. Polarization curves for all BMG alloys are similar. They exhibit low passive current density and very wide passive region (about 2.6 V) and the passive state can still be kept in even the potential exceeds 2 V (versus SCE). Furthermore, the electrochemical tests with stainless steel 304L were also performed in the H_2SO_4 solutions for comparison. It is found that the steel shows higher corrosion potential and passive current density, and narrower passive region than the ZrCu-based BMGs. The current density steeply increases at about 1.5 V (versus SCE), indicating that the passivation film is damaged and active dissolution occurs.

Fig. 2 shows the polarization curves of the BMG containing 0 and 4 at.% Nb in PBS (the dissolved-oxygen content in which was reduced to physiologically relevant value) open to air at 37 °C. It can be seen that both the Nb-bearing alloy's pitting potential and passive region are larger than Nb-free alloy, suggesting that the passive film formed upon the BMG containing Nb is more protective. After pitting, however, the alloy with 4 at.% Nb exhibits higher corrosion current density than Nb-free alloy, which represents that the corrosion reactions in Nb-bearing alloy are more severe at high potential.

To get further information on the chloride-induced localized corrosion mechanism, SEM was employed to investigate the morphologies of the samples after polarization tests in 3% NaCl. Fig. 3(a) and (c) shows the SEM images of the alloys containing zero and 4 at.% Nb, respectively, which were finally anodically polarized to about 40 mV below their pitting potentials in 3% NaCl solutions. It is found that some zones having dark contrast appear on the surface of Nb-free alloy (Fig. 3(a)) while only homogeneous surface can be observed from the surface of Nb-bearing alloy

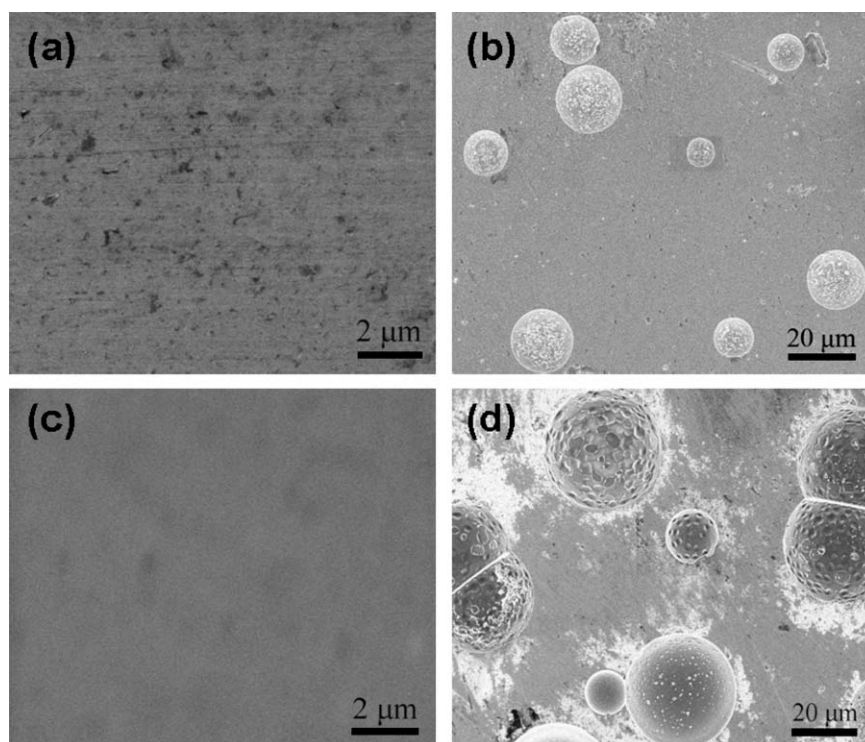


Fig. 3. SEM images taken from the corroded surface regions of $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ alloy with $x = 0$ and 4 at.% after anodic polarization in 3% NaCl solution: corrosion morphology of Nb-free alloy (a) and (b); corrosion morphology of alloy containing 4 at.% Nb (c) and (d).

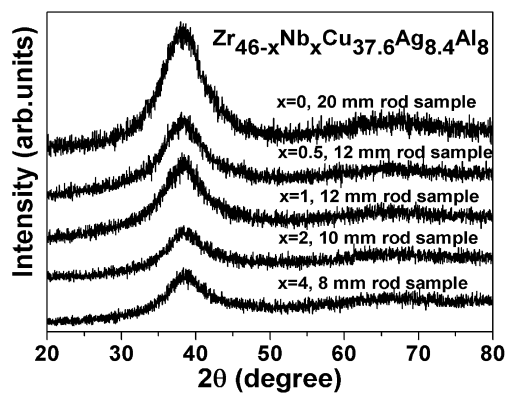


Fig. 4. XRD patterns for $Zr_{46-x}Nb_xCu_{37.6}Ag_{8.4}Al_8$ ($x = 0, 0.5, 1, 2$ and 4 at.%) bulk metallic glasses.

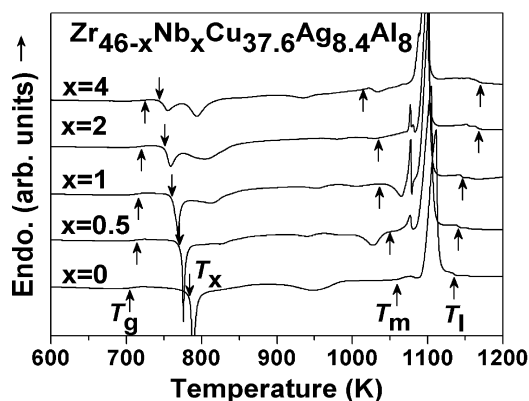


Fig. 5. DSC curves for $Zr_{46-x}Nb_xCu_{37.6}Ag_{8.4}Al_8$ ($x = 0, 0.5, 1, 2$ and 4 at.%) bulk metallic glasses.

(Fig. 3(c)) before pitting happens. The zones exhibiting dark contrast in Nb-free alloy are thought owing weak passive films, where will become the favorable sites for the adsorption and penetration of aggressive anions. In contrast, the surface on Nb-bearing alloy is still homogeneous although it was polarized to much higher potential than Nb-free alloy. This demonstrates that the passive film in Nb-free alloy already becomes unstable even though it was still in passive state, while the passive film in Nb-bearing alloy is more chemically homogeneous and chloride-resistant due to the modification of Nb. Fig. 3(b) and (d) exhibit the localized corrosion morphology of the two samples which were anodically polarized to the stage at which pitting corrosion just occurred. Some typical corrosion pits with different sizes distribute on the surface for both alloys. On the surface of Nb-bearing alloy, pits have higher density and larger sizes than those for Nb-free alloy in Fig. 3(b) and (d). The pitting potential of Nb-bearing alloy (-0.10 V) is much higher than Nb-free alloy. Once the passive film is broken and pitting is initiated, fast pit propagation into the reac-

tive metals occurs at the motivation of high potential which makes the pitting in Nb-bearing alloy more severe than Nb-free alloy. It is very interesting to find some micropores existing at the pits of Nb-bearing alloy. However, similar micropores were not found in Nb-free alloy.

3.2. Influence of Nb addition on GFA, thermal stability and mechanical property

Fig. 4 shows XRD patterns of as-cast $Zr_{46-x}Nb_xCu_{37.6}Ag_{8.4}Al_8$ ($x = 0, 0.5, 1, 2$ and 4 at.%) alloy rods with different diameters. All patterns exhibit broad diffraction peaks, revealing the amorphous nature for all the BMGs. Although GFA decreases with Nb addition, fully amorphous samples with critical diameters of at least 8 mm can still be produced by copper mould injection.

Fig. 5 shows DSC traces recorded at a constant heating rate of 0.33 K/s for all BMGs. They reveal glass transition followed by extended undercooled liquid region, exothermic crystallization peaks and melting endothermic peaks. Table 1 lists relevant thermal parameters: glass transition temperature (T_g), crystallization temperature (T_x), melting (T_m) and liquidus temperature (T_l), temperature range for the melting event, $\Delta T_m = T_l - T_m$, together with supercooled liquid region, $\Delta T = T_x - T_g$, reduced glass transition temperature $T_{rg} = T_g/T_l$ and γ value ($\gamma = T_x/(T_g + T_l)$) [20]. It is seen that Nb addition largely affects GFA from a diameter of above 20 mm (for Nb = 0) to about 8 mm (for Nb = 4 at.%). T_g increases while T_x decreases with the addition of Nb, which makes the supercooled liquid region steeply decrease from 73 K (for Nb = 0) to 18 K (for Nb = 4 at.%). The parameter T_{rg} keeps almost unchanged and γ value slightly decreases with the increase of Nb content. The parameter, ΔT , can roughly represent the GFA in the present system while the parameters, T_{rg} and γ , cannot.

The effect of Nb addition on the mechanical property was further investigated by performing compression tests at room temperature. The stress–strain traces for the $Zr_{46-x}Nb_xCu_{37.6}Ag_{8.4}Al_8$ ($x = 0, 0.5, 1, 2$ and 4 at.%) alloys are shown in Fig. 6. It is found that all the metallic glasses exhibit nearly the same yield strength of about 1900 MPa and about 1 – 3% plastic deformation after yielding. The compression plasticity exhibited no considerably enhancement due to the addition of Nb.

4. Discussion

4.1. Corrosion tests

As is observed from weight-loss measurements (see Fig. 1(a)–(c)), the corrosion–penetration rates are significantly decreased with the increase of Nb content. In polarization tests, it is also found that the pitting susceptibility was largely improved in 3% NaCl and PBS solutions and the corrosion potential was shifted to positive direction in 1 N HCl due to Nb addition. Qin et al. [16] reported that the addition of Nb in Cu–Hf–Ti BMGs can lead to the enrichment of not only Nb itself but also Hf and Ti (which are the most corrosion-resistant elements in the system), while depleting the element of Cu in passive film. Asami et al. [21] also

Table 1
The values of T_g , T_x , T_l , T_m , T_{rg} , γ , ΔT_m and ΔT for 2 mm diameter as-cast rods of $Zr_{46-x}Nb_xCu_{37.6}Ag_{8.4}Al_8$ ($x = 0, 0.5, 1, 2$ and 4 at.%) alloys prepared by copper mold injection.

$Zr_{46-x}Nb_xCu_{37.6}Ag_{8.4}Al_8$	Critical size (mm)	T_g (K)	T_x (K)	T_m (K)	T_l (K)	T_{rg} (K)	ΔT_m (K)	ΔT (K)	γ
$x = 0$	≥ 20	709 ± 2	782 ± 2	1060 ± 2	1136 ± 2	0.624	76	73	0.424
$x = 0.5$	≥ 12	716 ± 2	769 ± 2	1049 ± 2	1143 ± 2	0.626	94	53	0.414
$x = 1$	≥ 12	717 ± 2	760 ± 2	1030 ± 2	1147 ± 2	0.625	117	43	0.408
$x = 2$	10 – 12	719 ± 2	750 ± 2	1029 ± 2	1168 ± 2	0.616	139	31	0.398
$x = 4$	8 – 10	727 ± 2	745 ± 2	1013 ± 2	1170 ± 2	0.621	157	18	0.393

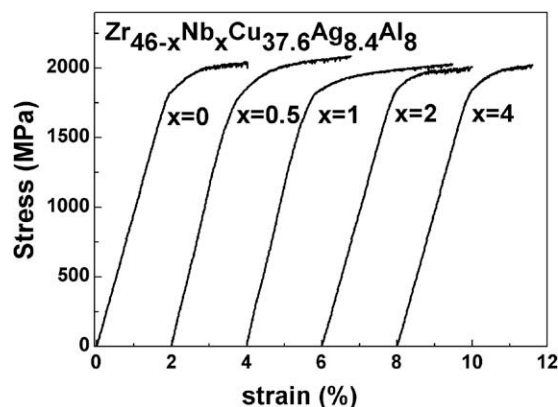


Fig. 6. Compressive stress-strain curves of as-cast $\text{Zr}_{46-x}\text{Nb}_x\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ ($x = 0, 0.5, 1, 2$ and 4 at.%) glassy rods with 2 mm in diameter and 4 mm in length at a strain rate of $4 \times 10^{-4} \text{ s}^{-1}$ at room temperature.

Table 2

Composition data at micropores and spots around micropores by EDX for $\text{Zr}_{42}\text{Nb}_4\text{Cu}_{37.6}\text{Ag}_{8.4}\text{Al}_8$ alloy after anodic polarization in 3% NaCl solution.

Position	Element concentration (± 1 at.%)					
	Zr	Cu	Ag	Al	Nb	O
Micropores	15	45	7	4	3	26
Spots around micropores	46	26	7	5	7	9

found Nb, Mo or Ta exhibited similar functions in Cu–Zr–Ti BMG system. This may also be the case in the present study, i.e. the property of the protective film was modified by the enrichment of Nb and Zr.

From the SEM results (see Fig. 3(a) and (c)), it can be clearly seen that although they were both in passive state, the corrosion morphology of Nb-free and Nb-bearing alloy in 3% NaCl was quite different. The surface of Nb-bearing alloy was still chemically homogeneous though it was polarized to higher potential, while some weak points already emerged on the surface of Nb-free alloy. This indicates that the property of passive film on the Nb-bearing alloy was considerably modified and became more dense and chloride-resistant. The micropores appeared at the pits of Nb-bearing alloy seems very curious. To understand the formation of micropores, we further performed composition analyzes at micropores and spots around micropores by EDX. Table 2 lists compositions at both places. It is clearly seen that at micropores Cu and O are enriched and Zr is depleted as compared to spots around micropores. Thus, we suggest that a decomposition of the amorphous matrix into two different amorphous phases (one is Cu enriched, the other is Zr and Nb enriched) could occur during the polarization tests in 3% NaCl solutions. Consequently, the phase enriched in Cu was selectively dissolved to form micropores at pits.

4.2. Influence of Nb addition on GFA, thermal stability and mechanical property

The glass-forming ability was decreased from above 20 to 8 mm with the increase of Nb content from 0 to 4 at.%. Why does the Nb addition deteriorate GFA and the thermal stability in the ZrCu-based BMG system? Two factors are considered to be the reason. First, Nb addition pushes compositions away from the eutectic point, which is confirmed by the enhancement of ΔT_m . Second,

Nb has positive heat of mixing (H_{mix}) with the main elements Zr, Cu and Ag [22]. The positive H_{mix} impedes interactions among the main components, consequently, increases the internal energy of the glassy alloy and finally deteriorates the thermal stability of the BMG system.

The novel ZrCu-based BMGs exhibited similar compression plasticity and the mechanical property got no obvious improvements by the micro alloying of Nb. Thus, it can be concluded that mechanical properties of the ZrCu-based BMGs have been not largely modified by addition of Nb.

5. Conclusions

Corrosion resistance in chloride-containing solutions for the newly developed ZrCu-based BMGs can be enhanced by microalloying of Nb. The alloys exhibit excellent corrosion behaviors in H_2SO_4 . The biocompatibility of the BMGs characterized by polarization tests in PBS is improved by Nb addition. The Nb-free alloy and alloy containing 4 at.% Nb undergo different pitting mechanisms in the NaCl solutions. Nb addition can deteriorate the GFA and thermal stability of ZrCu-based BMGs while almost no effect on the mechanical property of the alloys.

Acknowledgments

The authors would like to thank the financial support from the National Natural Science Foundation of China (Grants Nos. 50425102, 50601021, 50701038, 60776014, 60876002 and 10804096), Zhejiang University–Helmholtz cooperation fund, the Ministry of Education of China (Program for Changjiang Scholars and the Research Fund for the Doctoral Program of Higher Education), the Department of Science and Technology of Zhejiang province and Zhejiang University.

References

- [1] W.L. Johnson, MRS Bull. 24 (1999) 42.
- [2] A. Inoue, Acta Mater. 48 (2000) 279.
- [3] L.Y. Chen, Z.D. Fu, G.Q. Zhang, X.P. Hao, Q.K. Jiang, X.D. Wang, Q.P. Cao, H. Franz, Y.G. Liu, H.S. Xie, S.L. Zhang, B.Y. Wang, Y.W. Zeng, J.Z. Jiang, Phys. Rev. Lett. 100 (2008) 075501.
- [4] A. Peker, W.L. Johnson, Appl. Phys. Lett. 63 (1993) 2342.
- [5] A. Inoue, N. Nishiyama, H. Kimura, Mater. Trans.: JIM 38 (1997) 179.
- [6] F.Q. Guo, S.J. Poon, G.J. Shiflet, Appl. Phys. Lett. 83 (2003) 2575.
- [7] J. Schroers, W.L. Johnson, Appl. Phys. Lett. 84 (2004) 3666.
- [8] H. Ma, L.L. Shi, J. Xu, Y. Li, E. Ma, Appl. Phys. Lett. 87 (2005) 181915.
- [9] Q.K. Jiang, G.Q. Zhang, L. Yang, X.D. Wang, K. Saksl, H. Franz, R. Wunderlich, H. Fecht, J.Z. Jiang, Acta Mater. 55 (2007) 4409.
- [10] Q.K. Jiang, X.P. Nie, Y.G. Li, Y. Jin, Z.Y. Chang, X.M. Huang, J.Z. Jiang, J. Alloys Compd. 443 (2007) 191; Q. Zhang, W. Zhang, A. Inoue, Mater. Trans. 48 (2007) 629.
- [11] Q.K. Jiang, X.D. Wang, X.P. Nie, G.Q. Zhang, H. Ma, H.-J. Fecht, J. Bendnarcik, H. Franz, Y.G. Liu, Q.P. Cao, J.Z. Jiang, Acta Mater. 56 (2008) 1785.
- [12] L. Liu, Z. Liu, K.C. Chan, H.H. Luo, Q.Z. Cai, S.M. Zhang, Scripta Mater. 58 (2008) 231.
- [13] Q.K. Jiang, C.L. Qin, K. Amiya, S. Nagata, A. Inoue, R.T. Zheng, G.A. Cheng, X.P. Nie, J.Z. Jiang, Intermetallics 16 (2008) 225.
- [14] W.H. Jiang, F. Jiang, B.A. Green, F.X. Liu, P.K. Liaw, H. Choo, K.Q. Qiu, Appl. Phys. Lett. 91 (2007) 041904.
- [15] A. Gebert, U. Kuehn, S. Baunack, N. Mattern, L. Schultz, Mater. Sci. Eng. A 415 (2006) 242.
- [16] C.L. Qin, W. Zhang, K. Asami, H. Kimura, X.M. Wang, A. Inoue, Acta Mater. 54 (2006) 3713.
- [17] L. Liu, C.L. Qiu, H. Zou, K.C. Chan, J. Alloys Compd. 399 (2005) 144.
- [18] L.Q. Xing, Y. Li, K.T. Ramesh, Phys. Rev. B 64 (2001) 180211.
- [19] T.C. Huftagel, C. Fan, R.T. Ott, Intermetallics 10 (2002) 1163.
- [20] J. Das, M.B. Tang, K.B. Kim, R. Theissmann, F. Baier, W.H. Wang, J. Eckert, Phys. Rev. Lett. 94 (2005) 205501.
- [21] K. Asami, C.L. Qin, T. Zhang, A. Inoue, Mater. Sci. Eng. A 375 (2004) 759.
- [22] F.R. De Boer, R. Boom, W.C.M. Mattens, A.R. Miedema, A.K. Niessen, Cohesion in Metals, North-Holland Physics, Amsterdam, 1988.